

Noise-Aware Shrinkage for Differentially Private Zeroth-Order Fine-Tuning of Large Language Models

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Abstract

Differentially private zeroth-order optimization (DP-ZO) enables memory-efficient private fine-tuning of large language models using only forward evaluations. Existing aggregation-based DP-ZO methods reconstruct model updates at a fixed scale, ignoring that the strength of useful signals varies throughout training. Consequently, noise-dominated updates may receive excessive weight and degrade model utility. To address this issue, we propose SAGE, a noise-aware shrinkage method that adaptively attenuates privatized estimates according to their estimated signal quality. SAGE subtracts the known Gaussian noise variance from the observed second moment to estimate the underlying signal energy, stabilizes this estimate through temporal tracking, and compares its current signal-to-noise level with a warm-up reference to derive a bounded shrinkage factor. As pure post-processing, SAGE requires neither additional privacy budget nor model queries and introduces only constant additional state. Our theoretical analysis shows that shrinkage reduces the quadratic update-risk term faster than the linear descent term, preserving useful descent while limiting the influence of noise-dominated updates. Experiments on RoBERTa-large, OPT-1.3B, and OPT-6.7B demonstrate that SAGE outperforms existing baselines in most settings under the same privacy budgets while preserving the forward-only memory efficiency of DP-ZO.

Introduction

Fine-tuning large language models (LLMs) for specific tasks and domains has become a common practice in modern machine learning. Traditional fine-tuning relies on back-propagation and first-order optimizers, incurring substantial memory overhead from activations, gradients, and optimizer states (Kingma and Ba 2017; Loshchilov and Hutter 2019; Duchi, Hazan, and Singer 2011). To alleviate this bottleneck, MeZO introduces zeroth-order optimization (ZO) into LLM fine-tuning (Malladi et al. 2023). By estimating update directions through only two forward evaluations (Spall 1992) and regenerating random perturbations from their seeds, MeZO avoids backpropagation and reduces the training memory footprint to a level close to inference. When fine-tuning involves sensitive data, formal privacy protection is also essential. Building on this forward-only paradigm, differentially private zeroth-order optimization (DP-ZO) clips and perturbs the data-dependent directional response, enabling memory-efficient and privacy-preserving fine-tuning of LLMs (Tang

et al. 2025).

A central challenge in DP-ZO is the distortion introduced by clipping. Gradient scalar estimates can vary substantially across examples and random directions, making it difficult to select an appropriate fixed clipping threshold. A small threshold may excessively truncate informative signals, whereas a large threshold increases the sensitivity bound and consequently requires stronger privacy noise. DP-AggZO mitigates this issue by jointly clipping a K -dimensional vector of directional estimates before Gaussian perturbation (Bao et al. 2025). As K increases, the aggregate norm becomes more stable, reducing clipping error while preserving the forward-only and memory-efficient structure of DP-ZO.

Despite these advances, existing aggregation-based DP-ZO methods directly use the privatized gradient-scalar estimate vector obtained from multiple random directions, implicitly applying the same nominal scale across training steps. However, the energy of the underlying pre-noise aggregate can vary substantially during optimization (Asi et al. 2021; Zhang et al. 2024b; Moradi Kalarde et al. 2025), whereas the Gaussian noise variance is determined by the privacy mechanism rather than the current signal magnitude. Consequently, the effective signal-to-noise ratio of the released vector can be highly non-stationary. When the signal becomes weak, privacy noise accounts for a larger fraction of the update, increasing update risk and degrading optimization.

To examine this non-stationarity, we analyze the privatized gradient scalar estimates from a representative SNLI fine-tuning run. As shown in Figure 1, Gaussian privacy noise dominates the signal, with a median noise-to-signal energy ratio of 7.8. Here, reliability measures the estimated signal strength relative to the Gaussian privacy noise. It decreases from 0.213 at the end of warmup to approximately 0.100 in the late stage. These results show that signal quality varies during training, while fixed-scale updates use the same nominal scale across iterations. This mismatch motivates adaptive shrinkage. Scaling the privatized estimate vector by $m_t \in (0, 1]$ scales the useful descent term by m_t , but the quadratic update-risk term by m_t^2 . Thus, attenuating low-reliability estimates can suppress noise-induced risk more strongly than useful descent.

Based on the above observations, we propose **SAGE**, a noise-aware shrinkage method for adaptive update scaling in aggregation-based DP-ZO. Given a released privatized

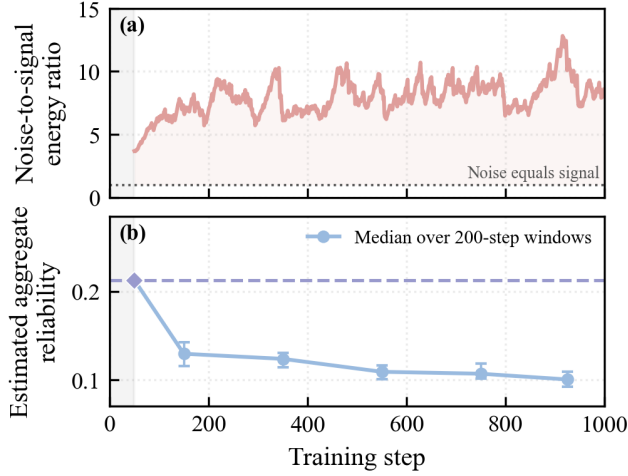


Figure 1: Non-stationary reliability of privatized gradient scalar estimates in DP-ZO fine-tuning. The signal quality varies throughout training under a fixed Gaussian noise level, while existing methods apply the same update scale regardless of estimate reliability.

gradient scalar estimate vector $\tilde{s}_t = u_t + \xi_t$, its observed energy $\|\tilde{s}_t\|_2^2/K$ contains an additive noise floor τ_t^2 determined by the Gaussian mechanism (Lee and Kifer 2018; Balle and Wang 2018; Tang and Lécuyer 2023; Tang, Shpilevskiy, and Lécuyer 2023). SAGE removes this known noise floor, tracks the corrected signal energy with a warmup-anchored exponential moving average, and computes a shrinkage factor $m_t \in (0, 1]$ to rescale the released estimate as $\bar{s}_t = m_t \tilde{s}_t$. SAGE never amplifies the estimate and, as pure post-processing, requires no additional queries or privacy cost. Its lightweight design can be seamlessly integrated into existing aggregation-based DP-ZO methods without modifying their optimization pipeline or privacy accounting. We apply SAGE to the privatized gradient scalar estimate vector and evaluate it on RoBERTa-large and OPT models across multiple text classification and natural language understanding tasks. The results show that SAGE improves private fine-tuning utility under identical privacy budgets and training settings.

Our contributions are summarized as follows:

- We identify and characterize the non-stationary reliability of Gaussian-privatized gradient scalar estimates, showing that fixed-scale updates do not account for changes in signal quality during training.
- We propose SAGE, a noise-aware shrinkage method that corrects the observed energy for the known Gaussian noise floor and adaptively scales privatized estimates through pure post-processing.
- We show theoretically that shrinkage suppresses the quadratic update-risk term more rapidly than the linear descent term, and empirically validate SAGE across RoBERTa-large and OPT models under the same privacy budgets and training settings.

Related Work

First-Order Differentially Private Fine-Tuning. First-order optimizers such as Adam, AdamW and AdaGrad (Kingma and Ba 2017; Loshchilov and Hutter 2019; Duchi, Hazan, and Singer 2011) are widely used for model training and fine-tuning. To protect sensitive training data, DP-SGD clips per-example gradients and injects Gaussian noise (Abadi et al. 2016), while DP-Adam and DP-AdamW provide differentially private variants of Adam and AdamW, respectively (Tang and Lécuyer 2023; Sun et al. 2025). Despite their strong utility, these methods require backpropagation, per-example gradients, and parameter-dimensional optimizer states, making their memory cost increasingly restrictive with growing LLM sizes.

Differentially Private Zeroth-Order Optimization. Following the SPSA framework (Spall 1992), MeZO (Malladi et al. 2023) reduces the memory overhead of LLM fine-tuning by replacing backpropagation with zeroth-order gradient estimates computed using only forward passes. For private fine-tuning, DPZero (Zhang et al. 2024a) and DP-ZO (Tang et al. 2025) privatize scalar ZO responses, while DP-AggZO (Bao et al. 2025) aggregates multiple independent ZO estimates to reduce the clipping error in DP-ZO and improve the accuracy-privacy tradeoff. However, existing methods still use privatized gradient scalar estimates at a fixed scale, so noise-dominated updates may receive excessive weight and degrade model utility.

Noise-Floor Correction under Differential Privacy. A related line corrects DP-induced bias in adaptive optimizers. DP-Adam (Tang and Lécuyer 2023) and DP-AdamBC (Tang, Shpilevskiy, and Lécuyer 2023) correct the second-moment estimator by subtracting the known Gaussian noise variance, while related noise-bias correction strategies are further explored in DP-FedAdamW (Liu et al. 2026) and FIBER (Dm et al. 2026). These methods correct per-coordinate first-order preconditioners, usually with $O(d)$ optimizer state.

Preliminaries

Differential privacy. Differential privacy (DP) provides a formal framework for limiting the privacy leakage of individual records. Two datasets D and D' are neighboring if one can be obtained from the other by adding or removing a single record. A randomized mechanism $\mathcal{M}$ satisfies (ϵ, δ) -differential privacy if, for all neighboring D, D' and any measurable output set $\mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in \mathcal{O}] \leq e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{O}] + \delta. \quad (1)$$

Zeroth-order and differentially private zeroth-order optimization. Zeroth-order (ZO) optimization estimates update directions using only function values (Liu et al. 2020). At step t , ZO samples a random perturbation direction $z_t \sim \mathcal{N}(0, I_d)$ and evaluates the loss at $\theta_t + \mu z_t$ and $\theta_t - \mu z_t$. For an example x , the resulting gradient scalar estimate is

$$\Delta_{z_t}(\theta_t; x) = \frac{\ell(\theta_t + \mu z_t; x) - \ell(\theta_t - \mu z_t; x)}{2\mu}, \quad (2)$$

where θ_t denotes the model parameters, μ is the perturbation scale, and $\ell(\theta; x)$ is the loss on example x .

Differentially private zeroth-order optimization (DP-ZO) clips and perturbs this data-dependent gradient scalar estimate before forming updates.

Differentially private aggregated zeroth-order optimization. Differentially Private Aggregated Zeroth-Order Optimization (DP-AggZO) extends DP-ZO by sampling K independent Gaussian directions $z_{t,1}, \dots, z_{t,K}$ and computing one gradient scalar estimate along each direction. Let $\Delta_{t,k}(x) = \Delta_{z_{t,k}}(\theta_t; x)$. For each example x , these estimates form the K -dimensional gradient scalar estimate vector

$$\Delta_{\text{agg}}(\theta_t; x) = \frac{1}{K} (\Delta_{t,1}(x), \dots, \Delta_{t,K}(x)). \quad (3)$$

Given a clipping threshold C , DP-AggZO jointly clips this vector as

$$\hat{\Delta}_{\text{agg}}(\theta_t; x) = \min \left(1, \frac{C}{\|\Delta_{\text{agg}}(\theta_t; x)\|_2} \right) \Delta_{\text{agg}}(\theta_t; x). \quad (4)$$

The clipped vectors are then aggregated over the sampled batch and perturbed with Gaussian noise to obtain the privatized gradient scalar estimate vector $\tilde{s}_t$.

Method

This section presents SAGE, a noise-aware shrinkage method for adaptive update scaling of privatized gradient scalar estimates. As illustrated in Figure 2, SAGE operates entirely on the released low-dimensional estimate vector.

Following DP-AggZO (Bao et al. 2025), at step t we obtain a privatized gradient scalar estimate vector

$$\tilde{s}_t = u_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \tau_t^2 I_K), \quad (5)$$

where $u_t \in \mathbb{R}^K$ contains the clipped and aggregated gradient scalar estimates before noise addition, and τ_t^2 is the known per-coordinate Gaussian noise variance. SAGE only post-processes $\tilde{s}_t$, leaving the queries, clipping rule, Gaussian mechanism, and privacy accountant unchanged.

Alg. 1 summarizes the resulting procedure. SAGE adds only noise-floor correction, scalar temporal tracking, and bounded shrinkage to the released estimate vector.

Estimating Reliability from Privatized Gradient Scalar Estimates

The key observation behind SAGE is that the reliability of the released aggregate is not stationary during training. Although the Gaussian noise variance is fixed by the privacy mechanism, the magnitude of the underlying optimization signal changes as optimization progresses. This motivates estimating the signal strength of the privatized gradient scalar estimates and adapting the update scale accordingly.

To estimate this reliability, SAGE exploits the known Gaussian noise structure of the released estimates. We first compute the squared magnitude of the released estimate vector:

$$r_t^2 = \frac{\|\tilde{s}_t\|_2^2}{K}. \quad (6)$$

The squared norm is used because the Gaussian perturbation introduces a predictable additive component into the

Algorithm 1: SAGE: Noise-Aware Shrinkage of Gradient Estimates

Input: Dataset D , steps T , sampling rate q , learning rate η , perturbation scale μ , noise multiplier σ , clipping threshold C , directions K , warmup W , EMA rate β , parameter λ , floor ρ , lower bound $m_{\min}$, Laplace scale ψ .

- 1: Release $\tilde{n} \leftarrow \max \{1, |D| + \text{Lap}(0, \psi)\}$, and set $\tilde{m} \leftarrow q\tilde{n}$.
- 2: Initialize buffers $\mathcal{B}_q, \mathcal{B}_\tau \leftarrow \emptyset$.
- 3: **for** $t = 1, \dots, T$ **do**
- 4: Sample batch B_t by Poisson sampling.
- 5: Form and privatize the K -dimensional gradient scalar estimate vector:

$$\tilde{s}_t \leftarrow \frac{1}{\tilde{m}} \left(\sum_{x \in B_t} \hat{\Delta}_{\text{agg}}(\theta_t; x) + \mathcal{N}(0, \sigma^2 C^2 I_K) \right).$$

- 6: Set $\tau_t^2 \leftarrow (\sigma C / \tilde{m})^2$.
- 7: Estimate debiased signal energy:

$$\hat{q}_t \leftarrow \max (\|\tilde{s}_t\|_2^2 / K - \tau_t^2, \rho).$$

- 8: **if** $t \leq W$ **then**
- 9: Append $\hat{q}_t$ to $\mathcal{B}_q$, append τ_t^2 to $\mathcal{B}_\tau$.
- 10: Set $m_t \leftarrow 1$.
- 11: **if** $t = W$ **then**
- 12: $v_{\text{ref}} \leftarrow \text{mean}(\mathcal{B}_q)$, $\tau_{\text{ref}}^2 \leftarrow \text{mean}(\mathcal{B}_\tau)$.
- 13: $v \leftarrow v_{\text{ref}}$, $\phi_{\text{ref}} \leftarrow v_{\text{ref}} / (v_{\text{ref}} + \lambda \tau_{\text{ref}}^2)$.
- 14: **end if**
- 15: **else**
- 16: $v \leftarrow (1 - \beta)v + \beta \hat{q}_t$.
- 17: $\phi_t \leftarrow v / (v + \lambda \tau_t^2)$.
- 18: $m_t \leftarrow \text{clip}(\phi_t / \phi_{\text{ref}}, m_{\min}, 1)$.
- 19: **end if**
- 20: Apply shrinkage: $\bar{s}_t \leftarrow m_t \tilde{s}_t$.
- 21: Update:

$$\theta_{t+1} \leftarrow \theta_t - \eta \sum_{k=1}^K \bar{s}_{t,k} z_{t,k}.$$

- 22: **end for**
- 23: **return** Model parameters θ .

observed second moment. Given

$$\tilde{s}_t = u_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \tau_t^2 I_K), \quad (7)$$

the conditional expectation satisfies

$$\begin{aligned} \mathbb{E}_\xi [r_t^2 \mid u_t] &= \frac{1}{K} \mathbb{E}_\xi [\|u_t + \xi_t\|_2^2 \mid u_t] \\ &= \frac{\|u_t\|_2^2}{K} + \tau_t^2. \end{aligned} \quad (8)$$

Therefore, the Gaussian mechanism contributes a known noise floor τ_t^2 to the observed estimate energy. SAGE removes this contribution and obtains a noise-floor-corrected estimate of the signal energy:

$$\hat{q}_t = \max (r_t^2 - \tau_t^2, \rho), \quad (9)$$

where $\rho > 0$ prevents degenerate estimates near the noise level.

Since $\hat{q}_t$ is computed from a single release, it can fluctuate substantially across iterations. SAGE therefore tracks the

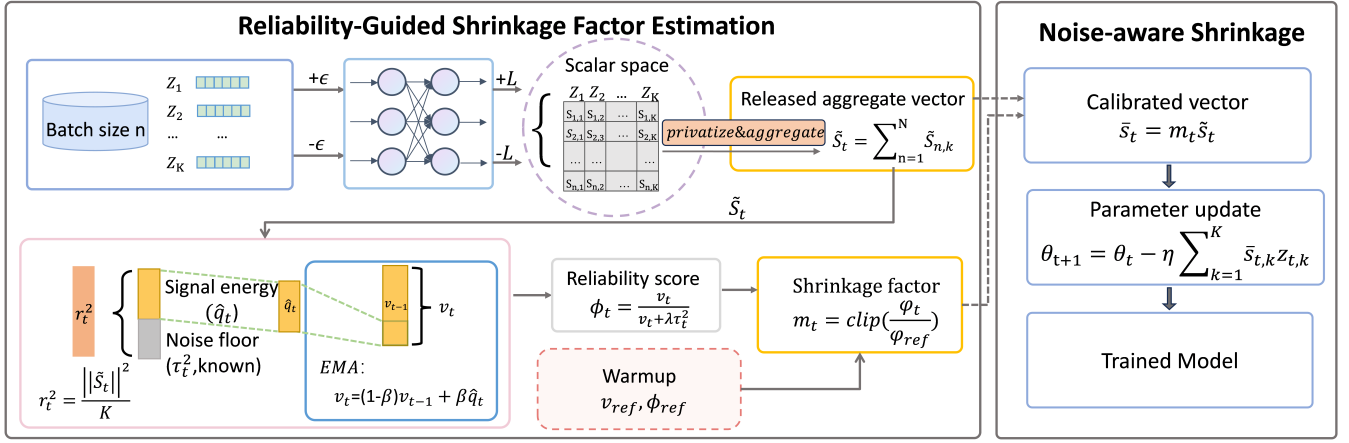


Figure 2: Overview of the SAGE framework. Starting from the released privatized gradient-scalar vector, SAGE explicitly corrects its observed energy for the known Gaussian noise floor, temporally smooths the resulting signal estimate, and converts it into a bounded, noise-aware shrinkage factor. This factor adaptively rescales the released vector before reconstructing the zeroth-order parameter update, effectively suppressing unreliable, noise-dominated updates without introducing additional model queries or privacy cost. The resulting update remains forward-only and memory-efficient.

temporal evolution of the estimated signal energy using an exponential moving average:

$$v_t = (1 - \beta)v_{t-1} + \beta\hat{q}_t. \quad (10)$$

This moving average only tracks the temporal variation of the reliability-related statistic and does not accumulate optimization directions. Therefore, it introduces only a scalar state and preserves the memory efficiency of DP-ZO.

During the first W steps, SAGE applies no shrinkage and collects:

$$v_{\text{ref}} = \frac{1}{W} \sum_{j=1}^W \hat{q}_j, \quad \tau_{\text{ref}}^2 = \frac{1}{W} \sum_{j=1}^W \tau_j^2. \quad (11)$$

The warmup statistics provide an early-training reference for reliability estimation. Instead of using an absolute signal threshold, SAGE measures the relative change of the current reliability with respect to this reference, allowing the shrinkage factor to adapt to the temporal variation of the privatized estimates.

Based on the tracked signal energy, SAGE constructs a bounded reliability score:

$$\phi_t = \frac{v_t}{v_t + \lambda \tau_t^2}, \quad (12)$$

where $\lambda > 0$ controls the contribution of the known noise level.

The reliability score follows an increasing-saturating form: it increases with the estimated signal energy and gradually approaches one when the released estimate becomes sufficiently reliable. However, directly using ϕ_t as the scaling factor would introduce a global dependence on the absolute magnitude of the reliability estimate, which varies across tasks, privacy settings, and optimization stages. SAGE therefore adopts a relative reliability measure with respect to the

early-training reference:

$$\phi_{\text{ref}} = \frac{v_{\text{ref}}}{v_{\text{ref}} + \lambda \tau_{\text{ref}}^2}. \quad (13)$$

The final scaling multiplier is obtained by comparing the current reliability with this reference:

$$m_t = \text{clip}\left(\frac{\phi_t}{\phi_{\text{ref}}}, m_{\text{min}}, 1\right). \quad (14)$$

The lower bound prevents excessive attenuation, while the upper bound ensures that SAGE never amplifies the released estimates. The released privatized estimate vector is then scaled as

$$\bar{s}_t = m_t \tilde{s}_t. \quad (15)$$

Since m_t depends only on privatized outputs and mechanism parameters, SAGE is pure post-processing and requires neither additional privacy budget nor forward queries.

Optimization Interpretation of Reliability-Aware Scaling

We analyze why attenuating unreliable privatized estimates can improve optimization. Consider the zeroth-order update after applying a scalar multiplier:

$$\Delta\theta_t = -\eta m_t (G_t + N_t), \quad (16)$$

where G_t denotes the clean random-direction ZO estimator and N_t denotes the update perturbation induced by Gaussian privacy noise after aggregation over random directions.

Under the standard smoothness assumption and ignoring higher-order finite-difference errors, the expected one-step loss change satisfies

$$\mathbb{E}[\Delta\mathcal{L}_t] \lesssim -\eta m_t S_t + \frac{L\eta^2 m_t^2}{2} (R_K S_t + N_K), \quad (17)$$

where $S_t = \|g_t\|_2^2$ denotes the underlying optimization-signal magnitude, R_K characterizes the variance introduced

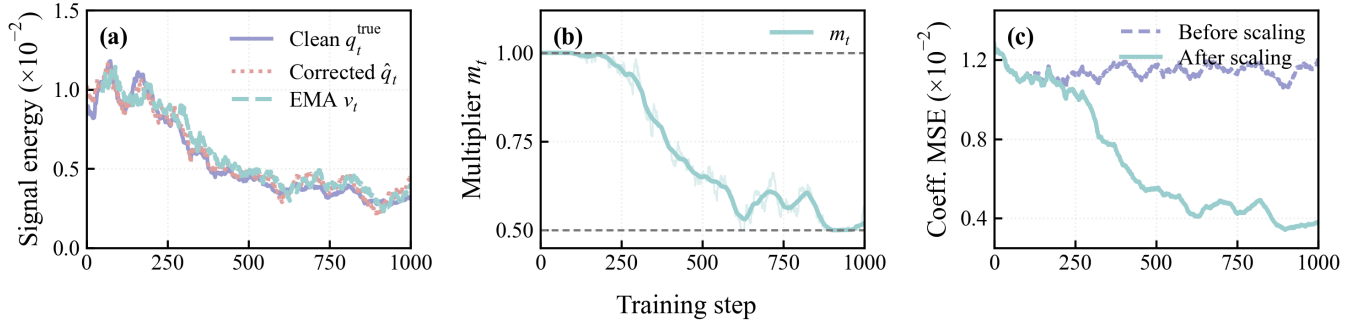


Figure 3: Diagnostics of SAGE on a representative RoBERTa-large SNLI run under $\epsilon = 6$ and $K = 64$. (a) The corrected statistic $\hat{q}_t$ and its EMA v_t track the clean aggregate energy $q_t^{\text{true}} = \|u_t\|_2^2/K$. (b) The multiplier m_t adapts within $[0.5, 1]$. (c) Adaptive scaling reduces the error to the clean aggregate by 42.3% after warmup and 65.5% over the final 200 steps. The clean aggregate u_t is used only for offline diagnostics and is never accessed by SAGE.

by random-direction estimation, and N_K represents the contribution of Gaussian privacy noise after random-direction aggregation.

The first term represents the expected descent contributed by the useful optimization signal, whereas the second term captures the update risk induced by random-direction estimation and privacy noise. Importantly, the two terms respond differently to scaling: the descent term decreases linearly with m_t , while the update-risk term decreases quadratically with m_t .

Ignoring the constraint $m_t \leq 1$, minimizing the one-step upper bound with respect to m_t gives the oracle scaling magnitude

$$m_t^* = \frac{S_t}{L\eta(R_K S_t + N_K)}. \quad (18)$$

This oracle quantity is unavailable because the clean signal and privacy perturbation are not separately observable, but it indicates that the preferred scale should increase with estimate reliability.

SAGE follows this principle using the computable score

$$\phi_t = \frac{v_t/\tau_t^2}{v_t/\tau_t^2 + \lambda}, \quad (19)$$

which is increasing and saturating in the estimated signal-to-noise level. The warmup-referenced multiplier therefore applies conservative attenuation only after the estimated reliability falls below its reference value.

Figure 3 examines estimator calibration, controller adaptation, and the effect of scaling. In panel (a), the corrected statistic closely tracks the temporal variation of the clean aggregate energy. For the displayed 50-step trajectories, its normalized mean absolute error and normalized root mean squared error are 7.3% and 7.9%, respectively, with a correlation of 0.987.

As the estimated reliability decreases, panel (b) shows that SAGE gradually reduces m_t rather than applying a fixed multiplier. Panel (c) compares

$$\frac{1}{K} \|\tilde{s}_t - u_t\|_2^2 \quad \text{and} \quad \frac{1}{K} \|m_t \tilde{s}_t - u_t\|_2^2, \quad (20)$$

showing that adaptive scaling reduces the post-warmup error by 42.3%, with a 65.5% reduction over the final 200 steps. Together, these diagnostics show that SAGE tracks reliability changes and improves the fidelity of the privatized aggregate in this representative run.

Experiments

We evaluate SAGE for differentially private full-parameter fine-tuning of RoBERTa-large, OPT-1.3B, and OPT-6.7B across text classification and question-answering tasks. We compare SAGE with representative first-order and zeroth-order private fine-tuning methods, using DP-AggZO as the primary baseline. For matched comparisons at a fixed number of directions K , DP-AggZO and SAGE use the same privacy budget, number of forward evaluations, training steps, and hyperparameter-tuning budget. Since SAGE operates entirely as post-processing of the privatized coefficient vector, it introduces no additional private queries or privacy cost. We further analyze the temporal behavior of aggregate reliability and characterize the computation and state overhead introduced by SAGE. All experiments are conducted on NVIDIA GeForce RTX 5090 GPUs.

Experimental Setup

Models and tasks. Following DP-AggZO, we evaluate full-parameter private fine-tuning on RoBERTa-large (355M) (Liu et al. 2019), OPT-1.3B, and OPT-6.7B (Zhang et al. 2022). RoBERTa-large is evaluated on SST-2, SST-5 (Socher et al. 2013), SNLI (Bowman et al. 2015), MNLI (Williams, Nangia, and Bowman 2018), RTE (Wang et al. 2018), and TREC (Li and Roth 2002), while OPT models are evaluated on SST-2 and SQuAD (Rajpurkar et al. 2016). We follow the same data splits and evaluation protocols as DP-AggZO, reporting accuracy for classification tasks and F1 score for SQuAD.

Baselines. We compare SAGE with DPZero, DP-AdamW, and DP-AggZO. DP-AggZO is the primary baseline since SAGE only performs post-processing on its privatized coefficient vector. Zero-shot evaluation is included as non-private references. All private results in Tables 1 and 2 are obtained

Privacy constraint	Algorithm	Sentiment		Natural Language Inference			Topic	Avg.
		SST-2	SST-5	SNLI	MNLI	RTE	TREC	Avg.
$\epsilon = 2, \delta = 10^{-5}$	DP-AdamW	91.3	47.4	74.3	73.6	72.5	91.1	75.03
	DPZero	92.3	46.6	73.4	62.2	70.2	83.3	71.33
	DP-AggZO ($K = 64$)	92.2	50.6	77.5	73.7	72.2	92.1	76.38
	SAGE ($K = 64$)	93.2	53.0	78.7	75.3	75.1	94.2	78.25
$\epsilon = 6, \delta = 10^{-5}$	DP-AdamW	92.3	48.8	81.4	76.4	77.1	92.3	78.05
	DPZero	91.8	48.9	77.4	67.5	71.5	87.2	74.05
	DP-AggZO ($K = 64$)	92.7	50.8	82.6	77.1	77.2	94.0	79.07
	SAGE ($K = 64$)	93.4	54.4	85.1	78.4	78.8	94.6	80.78
Perfect privacy	Zero-Shot	79.0	35.5	50.2	48.8	51.4	32.0	49.48

Table 1: Test performance on RoBERTa (355M). The best result is highlighted in bold.

Privacy constraint	Algorithm	SST-2		SQuAD	
		OPT-1.3B	OPT-6.7B	OPT-1.3B	OPT-6.7B
$\epsilon = 2, \delta = 10^{-5}$	DP-AdamW	91.2	OOM	77.7	OOM
	DPZero	87.1	92.2	72.0	78.1
	DP-AggZO ($K = 16$)	90.4	93.4	76.8	83.0
	DP-AggZO ($K = 64$)	90.9	94.1	78.8	83.2
	SAGE ($K = 16$)	91.1	94.1	77.9	83.9
	SAGE ($K = 64$)	91.5	94.4	80.1	84.6
$\epsilon = 6, \delta = 10^{-5}$	DP-AdamW	91.3	OOM	79.5	OOM
	DPZero	87.9	92.8	74.0	80.0
	DP-AggZO ($K = 16$)	91.1	94.5	77.2	83.1
	DP-AggZO ($K = 64$)	91.2	94.8	79.3	84.2
	SAGE ($K = 16$)	91.6	94.4	78.5	84.3
	SAGE ($K = 64$)	92.1	94.6	80.8	85.6
Perfect privacy	Zero-Shot	53.6	61.2	26.8	36.5

Table 2: Test performance on OPT-1.3B and OPT-6.7B. The best result is highlighted in bold. OOM stands for out of memory.

using our codebase. For matched comparisons, DP-AggZO and SAGE use the same privacy budget, query budget, number of training steps, and value of K . Their method-specific hyperparameter search spaces are reported in Supp. 3.

Training configurations. We consider (ϵ, δ) -DP with $\delta = 10^{-5}$ and $\epsilon \in \{2, 6\}$. Privacy accounting follows DP-AggZO using the Opacus RDP accountant. Unless otherwise specified, experiments use $T = 1000$ updates, target batch size 64, and perturbation scale $\mu = 10^{-3}$. We use $K = 64$ directions for RoBERTa-large and $K \in \{16, 64\}$ for OPT models.

SAGE uses fixed controller parameters $\beta = 0.03$, $W = 50$, $\lambda = 1$, $\rho = 10^{-8}$, and $m_{\min} = 0.5$ across all tasks.

Main Results

SAGE improves private fine-tuning on RoBERTa-large. Table 1 reports test accuracy on six classification benchmarks. SAGE achieves the highest average accuracy among all private methods under both privacy budgets, outperforming the first-order DP-AdamW, the single-direction DPZero, and the aggregation-based DP-AggZO.

Under $\epsilon = 6$, SAGE obtains an average accuracy of 80.78%, exceeding DP-AdamW, DPZero, and DP-AggZO by 2.73, 6.73, and 1.72 percentage points, respectively. In the matched comparison with DP-AggZO, SAGE improves all six tasks, with the largest gains on SST-5 and SNLI (+3.6 and +2.5 points). Under the stronger privacy constraint $\epsilon = 2$, SAGE again achieves the best private average of 78.25%, improving DP-AggZO by 1.87 points and outperforming it on all six benchmarks. The largest gains are observed on RTE, SST-5, and TREC, with improvements of 2.9, 2.4, and 2.1 percentage points, respectively. These improvements require neither additional private queries nor changes to the underlying privacy mechanism.

SAGE scales to larger language models. Table 2 evaluates SAGE on OPT-1.3B and OPT-6.7B for SST-2 and SQuAD. SAGE consistently outperforms DPZero across model sizes, tasks, and privacy budgets, achieving the best or near-best private performance. With $K = 64$ and $\epsilon = 2$, SAGE improves DP-AggZO on all four model-task combinations, with gains of 0.6 and 0.3 points on SST-2 and 1.3 and 1.4 points on SQuAD for OPT-1.3B and OPT-6.7B,

Method	SST-5	SNLI	MNLI	Avg.
SAGE	54.4	85.1	78.4	72.6
w/o noise correction	53.5	84.3	77.3	71.7
w/o EMA tracking	53.9	84.6	77.5	72.0
w/o warmup anchor	53.6	84.2	77.0	71.6

Table 3: Ablation study of noise correction and temporal tracking in SAGE on RoBERTa-large. Results are evaluated under $\epsilon = 6$ with $K = 64$.

Warmup steps	SST-5	SNLI	MNLI	Avg.
$W = 25$	54.0	84.7	77.8	72.2
$W = 50$ (default)	54.4	85.1	78.4	72.6
$W = 75$	54.3	84.9	78.1	72.4
$W = 100$	54.1	84.8	77.9	72.3

Table 4: Effect of warmup window in SAGE on RoBERTa-large. Results are evaluated under $\epsilon = 6$ with $K = 64$.

respectively.

Under $\epsilon = 6$, SAGE improves three of the four matched $K = 64$ settings, including gains of 1.5 and 1.4 F1 points on SQuAD, while remaining within 0.2 points of DP-AggZO on OPT-6.7B SST-2. It also surpasses DP-AdamW in all settings where the latter is executable, whereas DP-AdamW runs out of memory on OPT-6.7B. These results show that SAGE preserves the scalability of forward-only DP-ZO while improving the utility of the same privatized estimates.

Ablation Study

We conduct ablation studies on RoBERTa-large to investigate the contribution of each component in SAGE. All experiments are evaluated under $\epsilon = 6$ with $K = 64$. We consider three key components: (i) noise-floor correction, which removes the known Gaussian variance from the released aggregate energy; (ii) EMA tracking, which stabilizes the reliability estimate over training iterations; and (iii) warmup-based relative calibration, which provides a task-dependent reliability reference for shrinkage.

Effect of noise correction and temporal tracking. Table 3 shows the effect of removing individual components. Removing noise-floor correction consistently degrades performance, because the reliability estimator becomes biased by the additive privacy noise and tends to overestimate the signal strength. Removing EMA tracking also decreases utility, indicating that instantaneous reliability estimates contain substantial iteration-level fluctuations. The complete SAGE achieves the best average performance, demonstrating that accurate and stable reliability estimation is important for effective shrinkage.

Effect of warmup reference calibration. Table 4 studies the influence of the warmup window. Using a shorter warmup period provides a less stable reference due to insufficient statistics, whereas an overly long warmup delays the adaptation of the shrinkage controller. The default $W = 50$

Model	Dataset	Algorithm	Memory
RoBERTa	SST-2	DP-AdamW	11.38 GiB
		DP-AggZO	2.99 GiB
		SAGE	2.99 GiB
RoBERTa	MNLI	DP-AdamW	12.35 GiB
		DP-AggZO	3.25 GiB
		SAGE	3.25 GiB
RoBERTa	SNLI	DP-AdamW	11.39 GiB
		DP-AggZO	2.99 GiB
		SAGE	2.99 GiB

Table 5: Peak memory usage of DP-AdamW, DP-AggZO and SAGE, tested on a 5090 GPU. The memory usage of DP-AggZO with different K values is identical.

achieves the best overall performance, showing that a moderate warmup period provides a reliable balance between reference stability and adaptation speed.

Computation and Memory Efficiency

SAGE preserves the query and state efficiency of DP-AggZO. DP-AggZO and SAGE require exactly the same $2K$ forward evaluations per parameter update. SAGE adds only coefficient-space post-processing: computing $\|\tilde{s}_t\|_2^2$, subtracting the known Gaussian noise floor, updating a scalar EMA state, evaluating the shrinkage multiplier, and rescaling the K -dimensional coefficient vector. These additional operations require only $O(K)$ arithmetic per step and no model evaluations.

The persistent state introduced by SAGE consists of a constant number of scalar statistics, including v_t , v_{ref} , τ_{ref}^2 , and ϕ_{ref} . It therefore requires $O(1)$ additional optimizer state and introduces no parameter-dimensional gradient, momentum, or second-moment buffers. SAGE consequently preserves the forward-only and memory-efficient structure of DP-AggZO while providing reliability-aware post-processing of its privatized coefficient aggregate. Table 5 confirms no extra memory cost for SAGE.

Conclusion

In this work, we propose SAGE, a noise-aware shrinkage method for differentially private zeroth-order fine-tuning of large language models. SAGE estimates the time-varying signal quality of privatized zeroth-order coefficients by correcting their observed energy for the known Gaussian noise floor, and then adaptively scales their contribution through bounded post-processing. Our analysis shows that such shrinkage reduces the quadratic update-risk term more strongly than the useful descent term, providing a principled explanation for attenuating noise-dominated updates. Experiments across multiple language models and downstream tasks demonstrate that SAGE consistently improves the utility of DP-ZO fine-tuning under matched privacy budgets, query budgets, and training settings, while preserving its forward-only and memory-efficient structure.

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